ECE 340 Exam 1

Instructional Objectives

1. Outline the classification of solids as metals, semiconductors, and insulators and distinguish direct and indirect semiconductors. (1)
2. Determine relative magnitudes of the effective mass of electrons and holes from an E(k) diagram. (1)
3. Calculate the carrier concentration in intrinsic semiconductors. (1)
4. Apply the Fermi-Dirac distribution function to determine the occupation of electron and hole states in a semiconductor. (1)
5. Calculate the electron and hole concentrations if the Fermi level is given; determine the Fermi level in a semiconductor if the carrier concentration is given. (1)
6. Determine the variation of electron and hole mobility in a semiconductor with temperature, impurity concentration, and electrical field. (1)
7. Apply the concept of compensation and space charge neutrality to calculate the electron and hole concentrations in compensated semiconductor samples. (1)
8. Determine the current density and resistivity from given carrier densities and mobilites. (1)
9. Calculate the recombination characteristics and excess carrier concentrations as a function of time for both low level and high level injection conditions in a semiconductor. (1)
10. Use quasi-Fermi levels to calculate the non-equilibrium concentrations of electrons and holes in a semiconductor under uniform photoexcitation. (1)
11. Calculate the drift and diffusion components of electron and hole currents. (1)
12. Calculate the diffusion coefficients from given values of carrier mobility through the Einstein's relationship and determine the built-in field in a non-uniformly doped sample. (1)
13. Calculate diffusion current for steady-state carrier injection. (1)

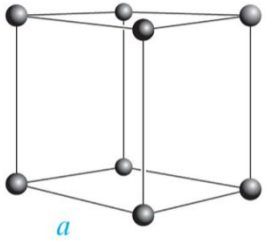
Course overview/Tips

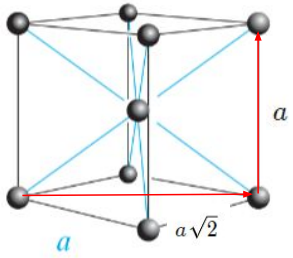
- Unit analysis
- Try to memorize certain quantities ahead of time (bandgaps, kT/q, n_i)
- Try to memorize equations (at least what they look like) ahead of time
- The time crunch is **REAL**
- Do not let this course discourage you from continuing study in semiconductors

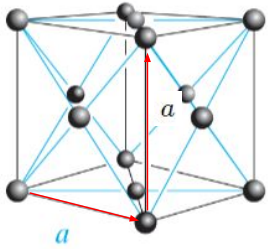
Crystal Lattice

- Diamond vs Zinc Blende
- Silicon is Diamond
- Categories of Cubic Lattices
  - Simple Cubic
  - Body Centered Cubic (BCC)
  - Face Centered Cubic (FCC)

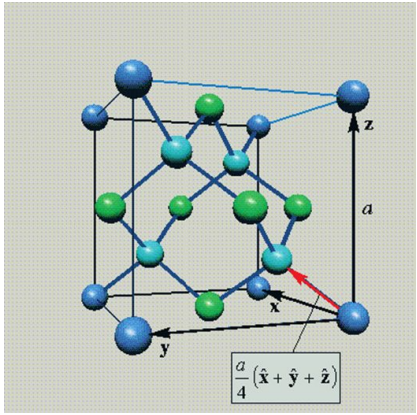
Cubic Lattice Structures







Structure	Simple Cubic	Body Centered Cubic (BCC)	Face Centered Cubic (FCC)
Length to Nearest Neighbor	a	$l = \frac{1}{2}\sqrt{(a\sqrt{2})^2 + a^2} = \frac{\sqrt{3}}{2}a$	$l = \frac{1}{2}\sqrt{a^2 + a^2} = \frac{\sqrt{2}}{2}a$
Atoms per Lattice	1	2	4



Diamond Lattice

Lattice Practice Problem

Find the atomic packing factor (APF) of a diamond cubic lattice. The atomic packing factor (APF) of a crystal structure is the fraction of the unit cell volume filled by atoms. Assume each atom is a hard sphere and nearest neighbors are touching.

Lattice Practice Problem Answer

Finally, for diamond we only need to look at the atom at the "origin" and its relation to the nearest neighbor. In a diamond lattice, you have two interspersed FCC lattices with one being located $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ away from the other. This means that the atom at (0, 0, 0) is bonded to the atom at $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, and we can extract the radius of each atom from our hard-sphere assumption.

$$2r = \sqrt{(\frac{a}{4})^2 + (\frac{a}{4})^2 + (\frac{a}{4})^2} = \frac{a\sqrt{3}}{4} \rightarrow r = \frac{a\sqrt{3}}{8}$$

Then, we need to find the APF from:

$$\text{APF} = \frac{\text{volume of atoms}}{\text{volume of unit cell}}$$

The number of atoms in a unit cell is calculated from number of lattice points multiplied by the number of atoms in the basis. The number of lattice points in the unit cell comes from 8 lattice points on the corners and 6 lattice points on the faces. $8 * (\frac{1}{8}) + 6 * (\frac{1}{2}) = 4$ lattice points in the unit cell. However, each lattice point has 2 atoms in it from the basis, which means diamond has 8 atoms in the unit cell. Putting this all together yields:

$$\text{APF} = \frac{8 * 4\pi(\frac{a\sqrt{3}}{8})^3}{3} \div a^3 = \frac{\pi\sqrt{3}}{16}$$

Miller Indices

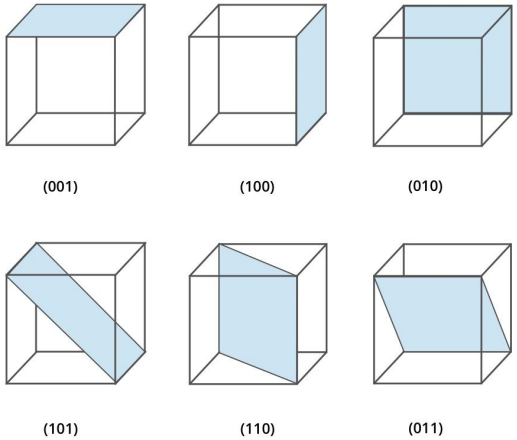
Directions: [hkl] or <hkl> family

1. Determine the unit vector

Planes: (hkl) or {hkl} family

1. Move plane off origin
2. Determine the axis intercepts
3. Take the reciprocal

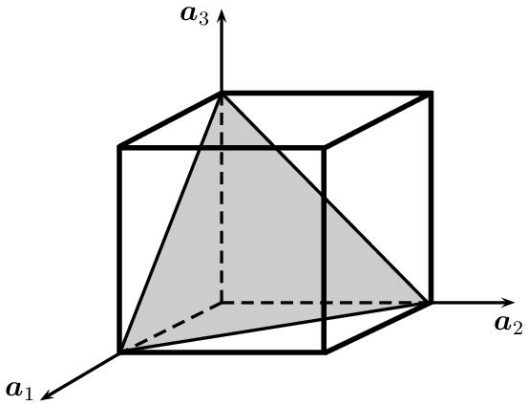
Negatives use an overline



$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Miller Indices Question

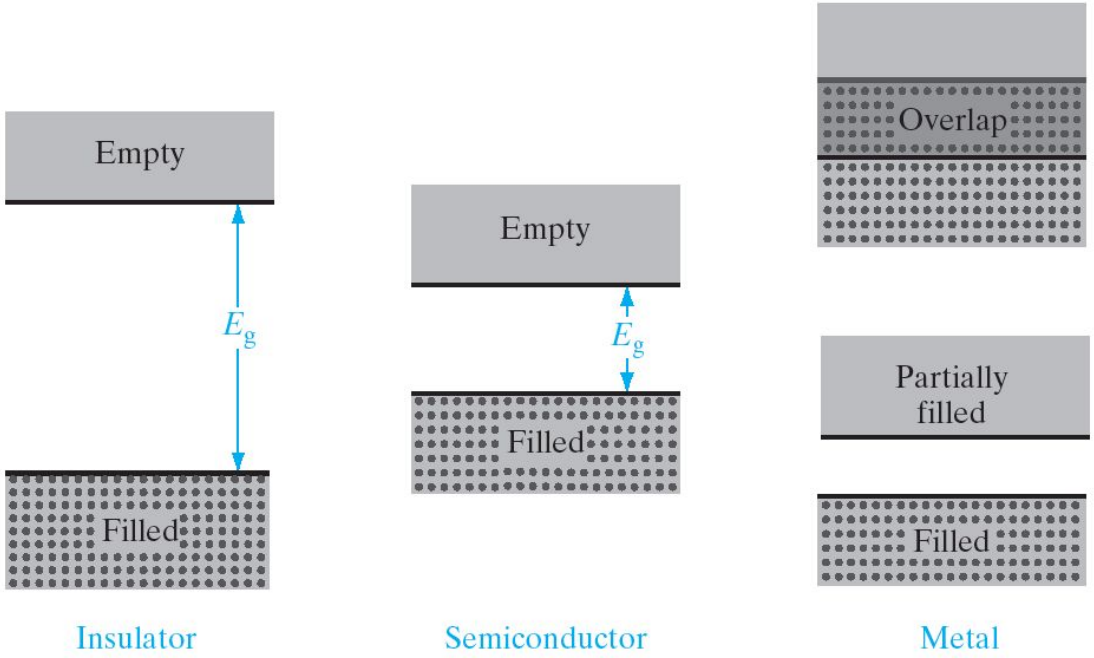
Identify the following plane:



It's the (111) plane!

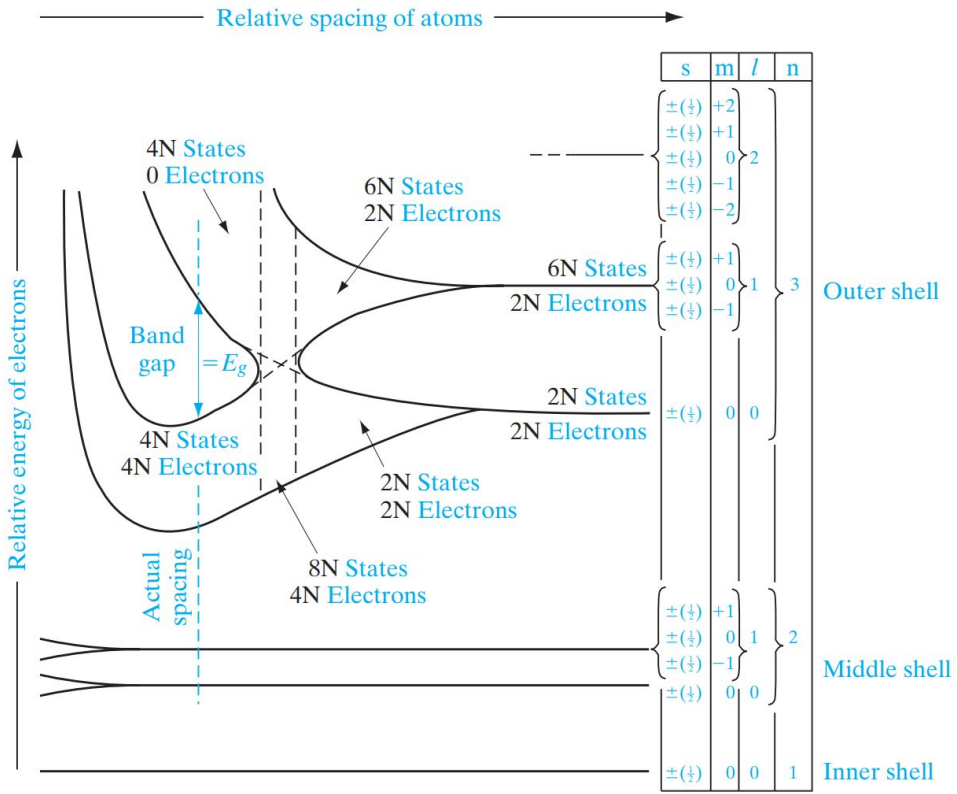
What is a semiconductor?

What temperature are these materials at?

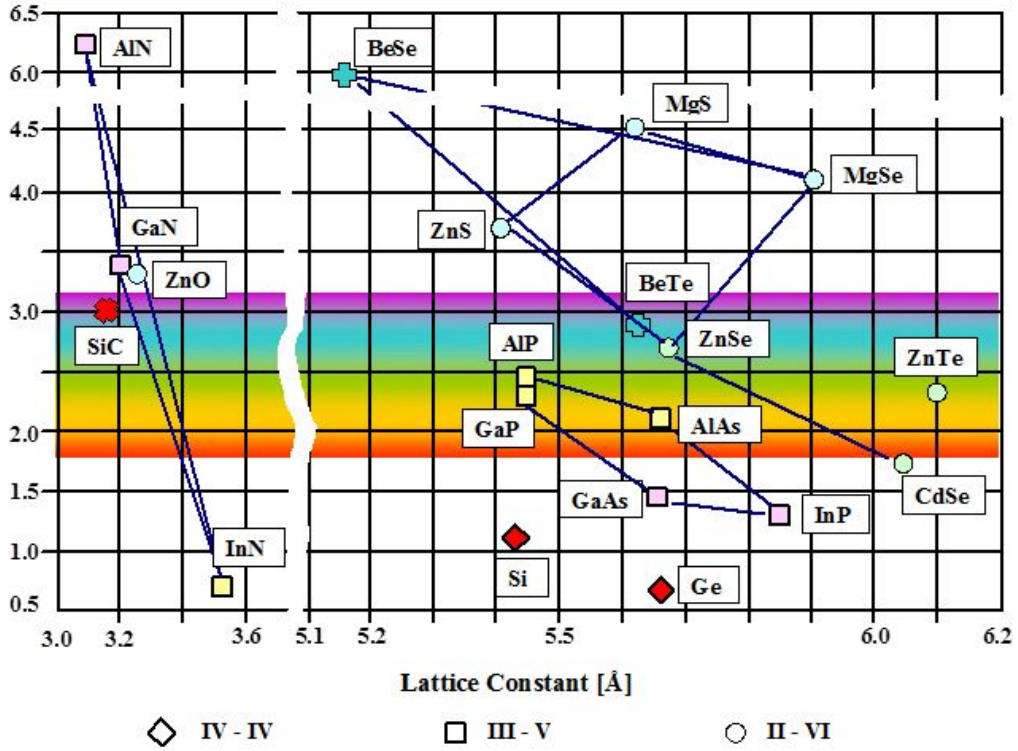


What is a bandgap?

- Quantum physical coupling
  - Removes degeneracy
- The closer the atoms, the further the energy levels split

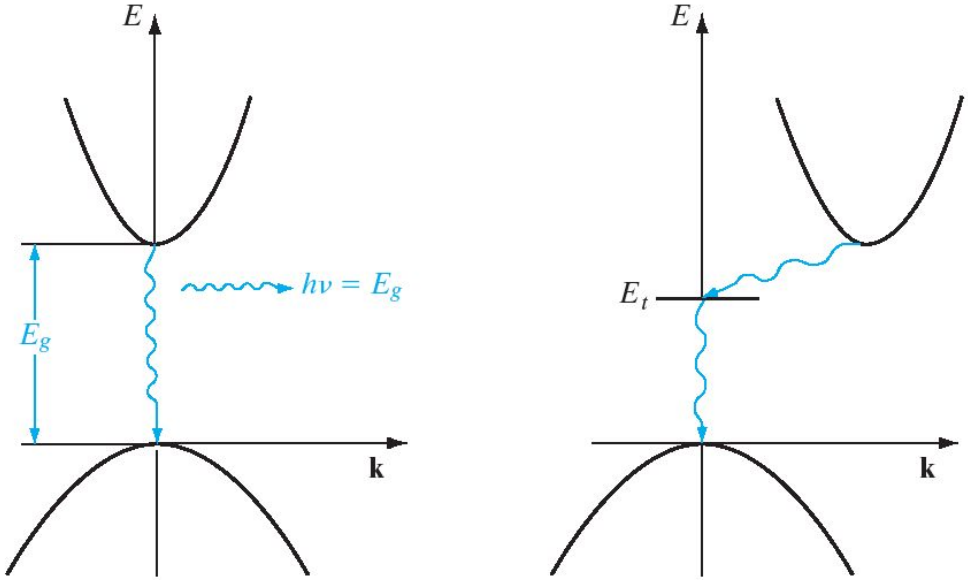


Structure vs Bandgap



- Lattice Matching
- Mix materials to fine-tune parameters

Direct vs Indirect Semiconductors



(a) Direct

(b) Indirect

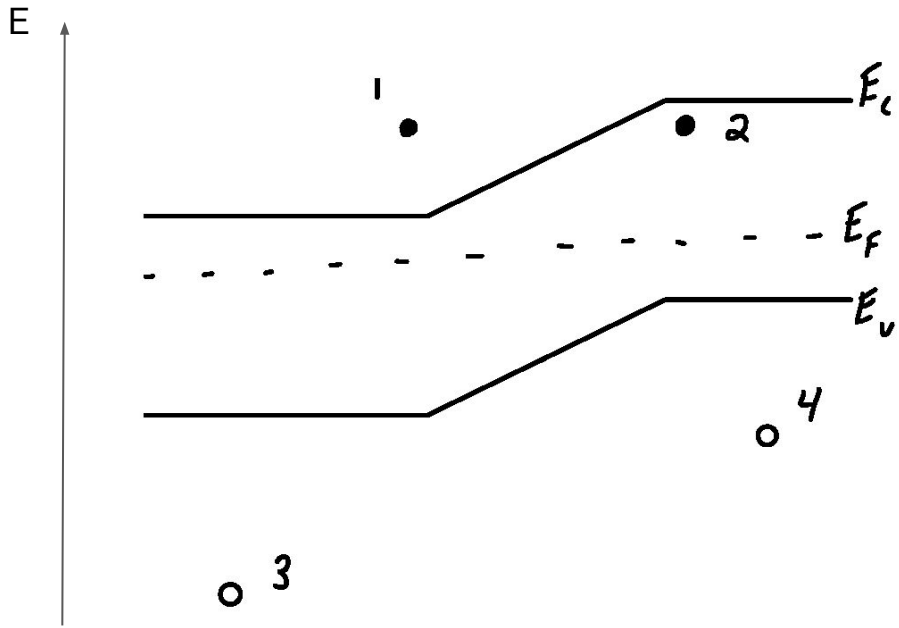
Charge Carriers in a Semiconductor

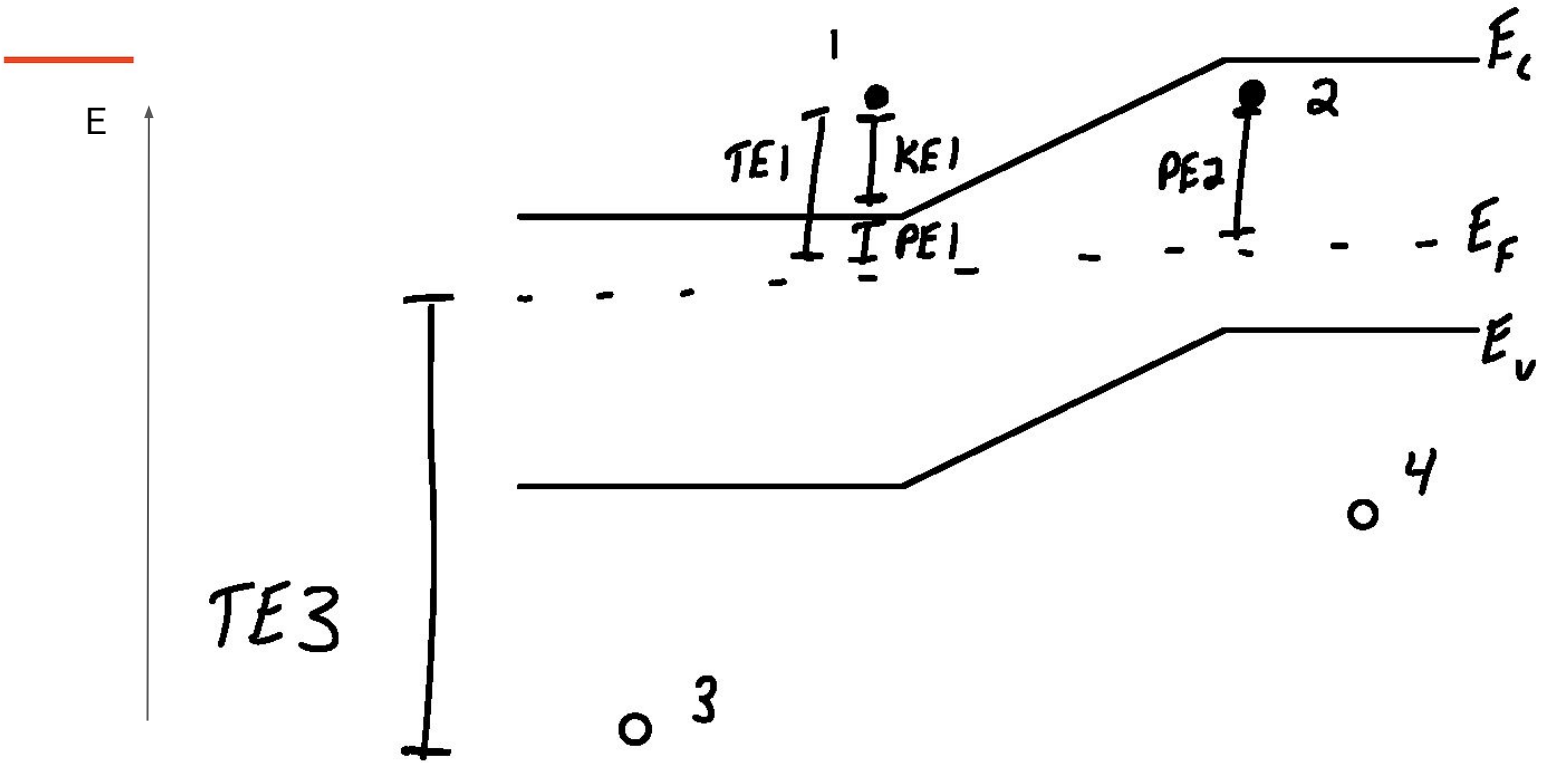
Electrons (-)
on top

Holes (+)
on bottom

Which hole (3 or 4) has more potential energy?

Which carrier (out of all) has the most total energy?



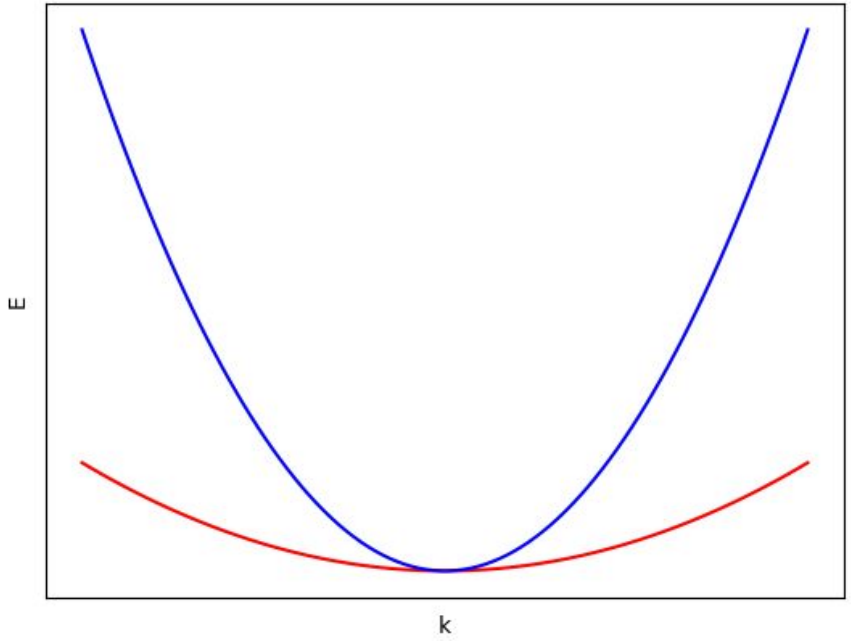

E
1
2
E_c
TE1
KE1
PE2
PE1
E_f
E_v
4
TE3
3

Effective Mass

$$m^* = \frac{\hbar^2}{d^2E/d\mathbf{k}^2}$$

Wider parabola -> Lower second derivative -> Higher effective mass

Which has a higher effective mass?



Fermi Dirac Distribution

- Gives the probability an energy state E will be occupied by an electron

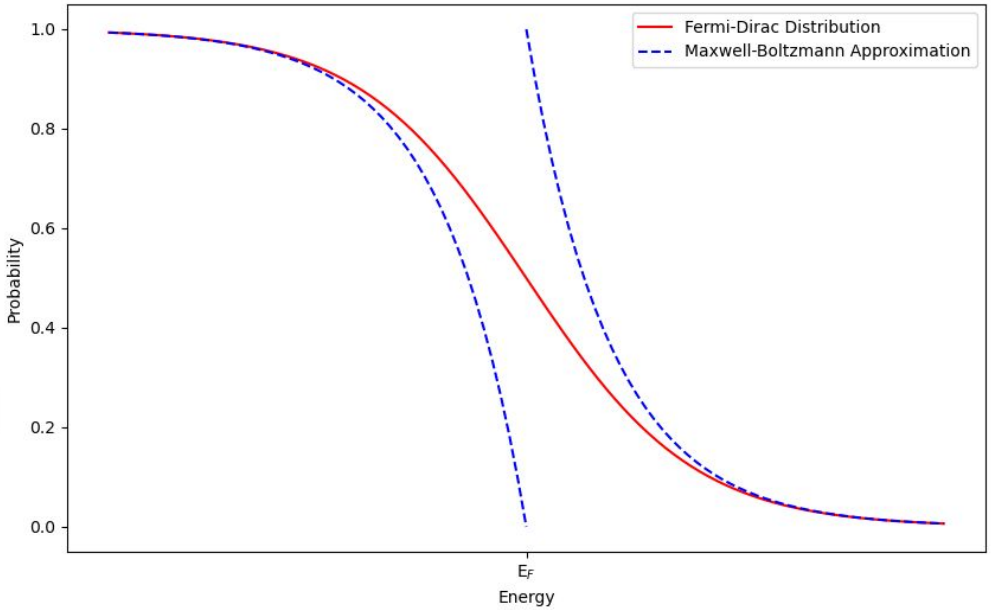
$$f(E) = \frac{1}{1 + e^{(E-E_F)/kT}}$$

- The Fermi level is the energy E where the probability of occupation by an electron is ½
- Fermi level at approximately midgap for intrinsic semiconductors
- At 0K: all states above E_F are empty and all states below are full

Maxwell-Boltzmann Approximation

If the energy level(s) you care about are sufficiently far enough away from the fermi level:

$$f(E) = \frac{1}{1 + exp[\frac{E - E_F}{kT}]} \approx exp[\frac{E_F - E}{kT}]$$

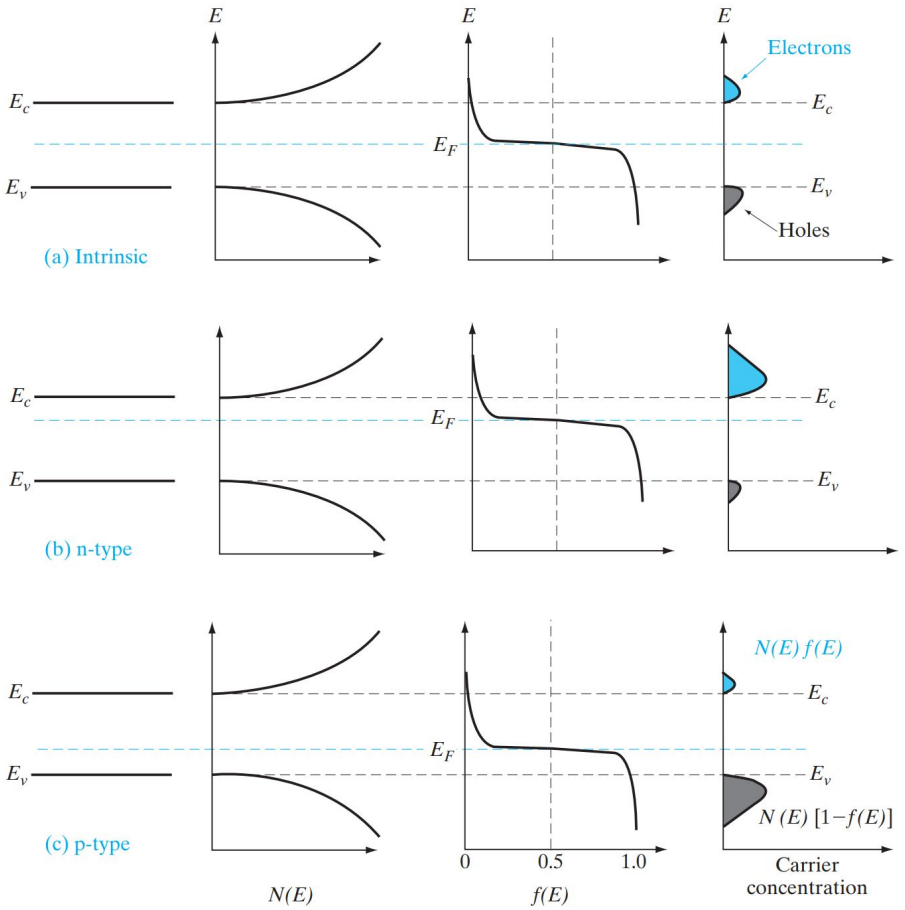


Density of States

D.O.S. is proportional to square root of energy

- Appendix IV for derivation

Remember, exponential decay beats square root growth



Equilibrium Carrier Concentration

$$n_0 = \int_{E_C}^{\infty} f(E)N(E)dE$$

$$p_0 = \int_{-\infty}^{E_V} [1 - f(E)]N(E)dE$$

$$n_0 = n_i e^{(E_F - E_i)/kT}$$

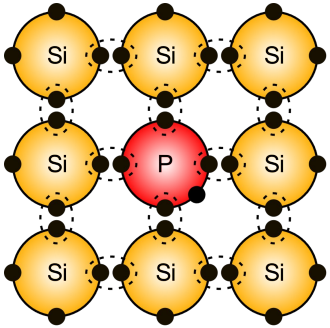
$$p_0 = n_i e^{(E_i - E_F)/kT}$$

$$n_0 p_0 = n_i^2$$

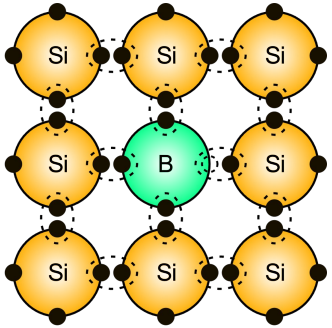
$$\frac{dE_F}{dx} = 0$$

Extrinsic Semiconductors

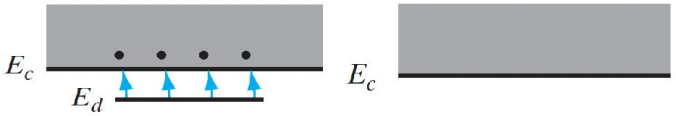
Donors -> Donate electrons to the conduction band
Acceptors -> Accept electrons from valence band (leaving a hole)



n-type

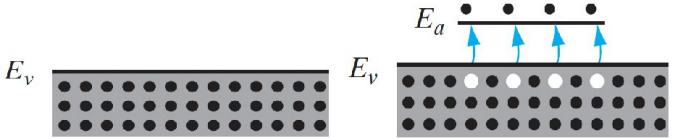


p-type



Donation of electron from the donor level to the Conduction band

Acceptance of electron from the valence band to the Acceptor energy level



Space Charge Neutrality

- When $N_d >> n_i$: $N_d \cong n_0$
- When $N_a >> n_i$: $N_a \cong p_0$

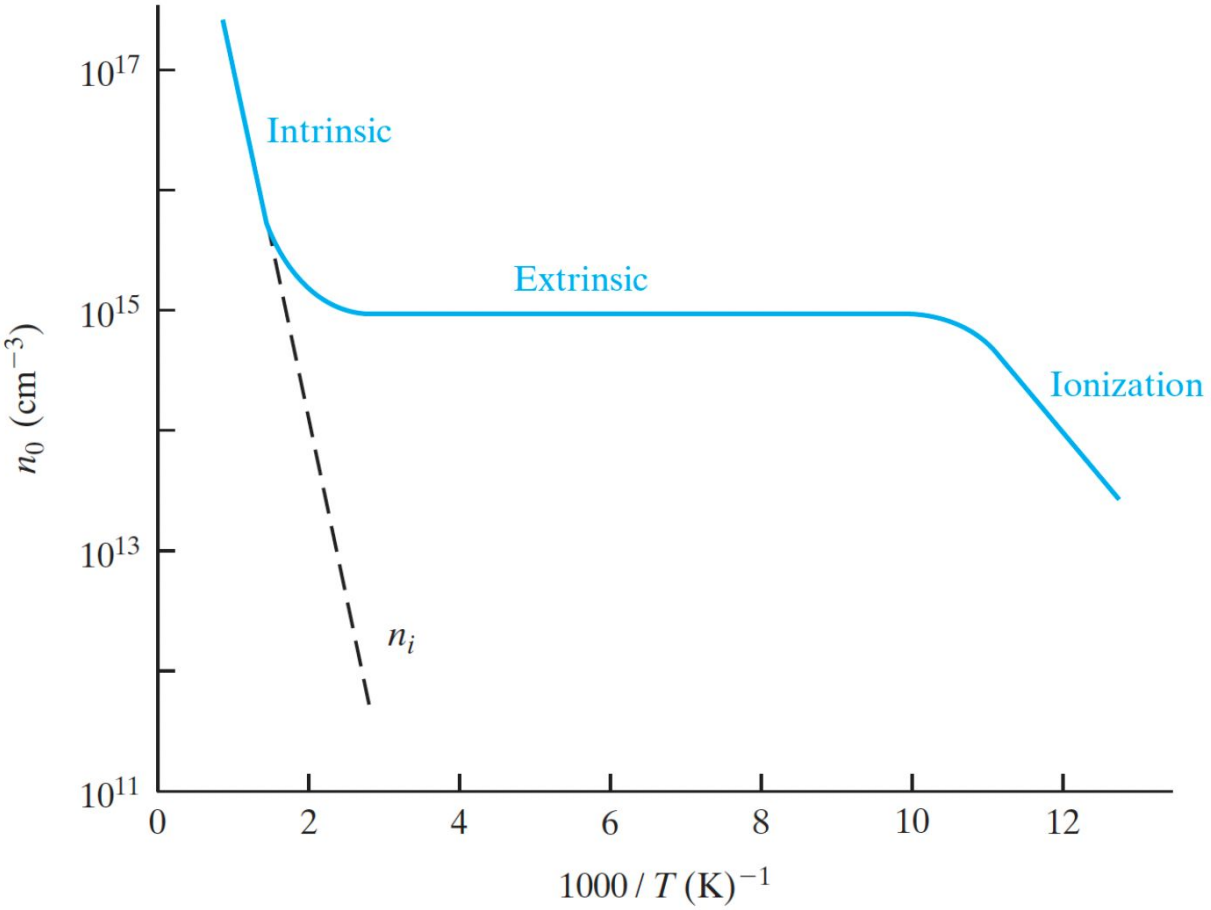
- **Compensation Doping**: (3 cases)
- When $N_d - N_a >> n_i$: $N_d - N_a \cong n_0$
- When $N_a - N_d >> n_i$: $N_a - N_d \cong p_0$
- When $|N_d - N_a| \approx n_i$: use quadratic formula (easy to derive)

Compensation: Intrinsic Condition when Electrostatically neutral.

$$n - p + N_A^- - N_D^+ = 0$$

$$n = \frac{N_D - N_A + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2}$$

$$p = \frac{-(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2}$$


Intrinsic
Extrinsic
Ionization
n_i
10^{17}
10^{15}
10^{13}
10^{11}
n_0 (cm^{-3})
0
2
4
6
8
10
12
1000 / T (K)$^{-1}$

Question

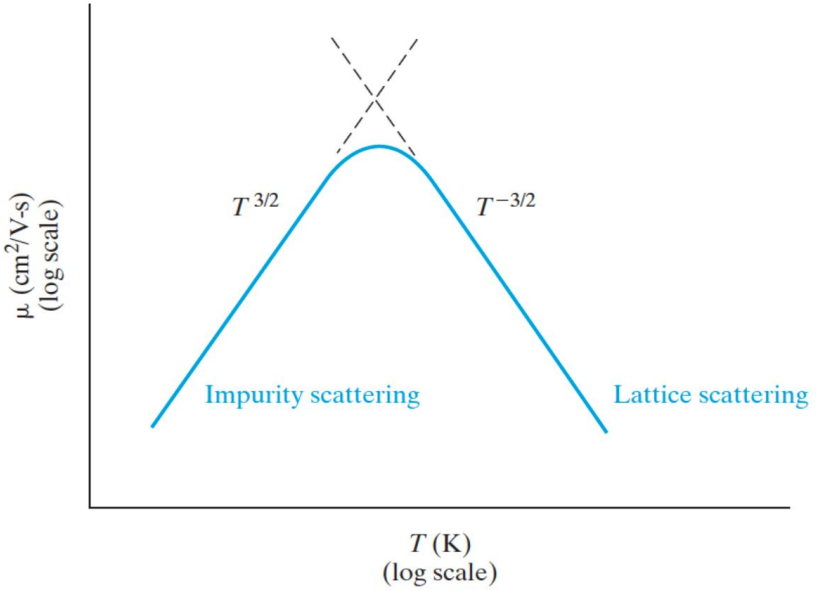
Assume you start with a doped semiconductor. How will the Fermi level change as you increase temperature?

Mobility

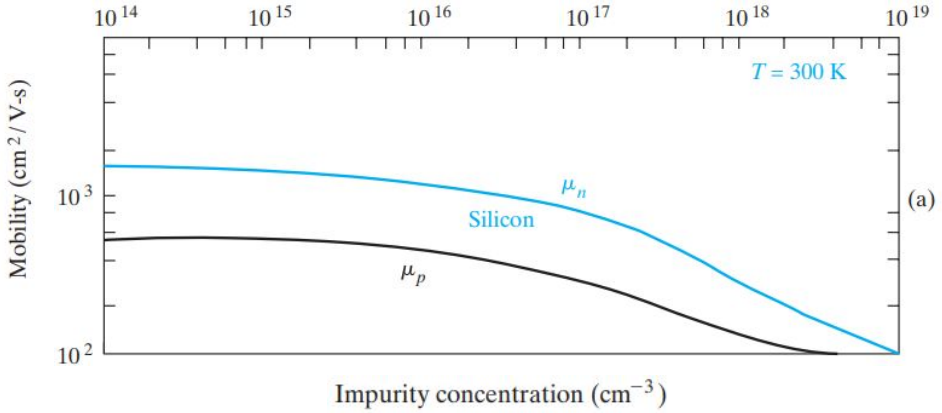
- The ease through which carriers drift through a material
- Units [cm^2/Vs]
- Add them as parallel resistors (smallest dominates)

$$\mu_n \equiv \frac{q\bar{t}}{m_n^*} \qquad \mu_n = -\frac{\langle v_x \rangle}{\mathscr{E}_x}$$

$$\frac{1}{\mu_{Tot}} = \frac{1}{\mu_{impurity}} + \frac{1}{\mu_{lattice}} + \frac{1}{\mu_3} + \dots$$



Reminder: Log Graphs



Halfway point on a log graph is 10^(½) = 3.162, NOT 5

5 is at around 7

Mobility vs Impurity is *total* impurities, Na + Nd

Conductivity

- Conductivity: $\sigma = q(n\mu_n + p\mu_p)$
  - Units of [S/m]
- Resistivity: $\rho = 1/\sigma$
  - Units of [Ω/m]
- Resistance: $R = \rho L/A = L/wt\sigma$

Diffusion Coefficient

- How good something is at diffusing
- Units $[cm^2/s]$

Einstein Relation

$$\frac{D}{\mu} = \frac{kT}{q}$$

Drift and Diffusion

- Drift – Carrier movement due to an electric field
- Diffusion – Carrier movement due to a concentration gradient
  - Random motion

$$J_n(x) = q\mu_n n(x)\mathcal{E}(x) + qD_n\frac{dn(x)}{dx}$$

drift diffusion

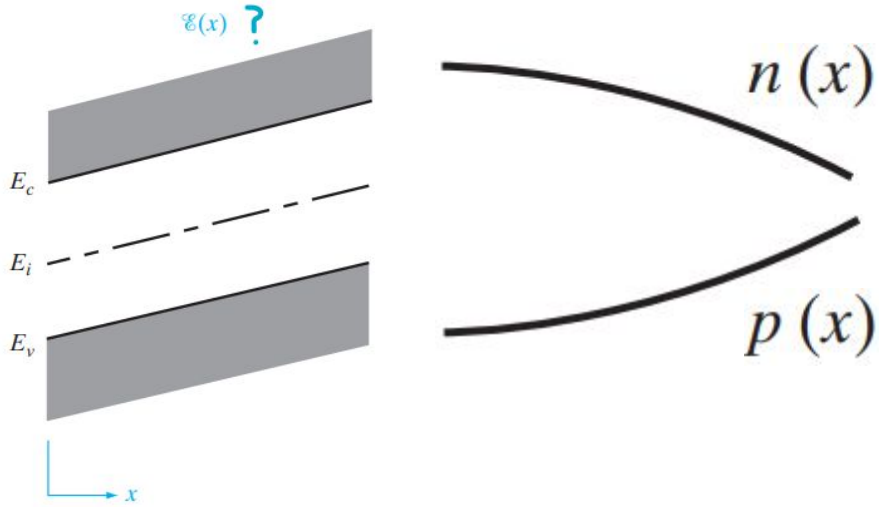
Conductivity σ
[S/cm or 1/Ohm cm]

$$J_p(x) = q\mu_p p(x)\mathcal{E}(x) - qD_p\frac{dp(x)}{dx}$$

$$J(x) = J_n(x) + J_p(x)$$

Drift and Diffusion

- Fill out the following chart given this information:



ϕn(drift)	Jn(drift)
ϕn(diff.)	Jn(diff.)
ϕp(drift)	Jp(drift)
ϕp(diff.)	Jp(diff.)

Assume equal magnitude for Jdrift and Jn, what is Jtot (for each carrier)?

Carrier Lifetime

$$\frac{dn(t)}{dt} = \alpha_r n_i^2 - \alpha_r n(t)p(t)$$

Equilibrium change in electron concentration is: thermally generated carriers – recombined carriers

$$\frac{d\delta n(t)}{dt} = \alpha_r n_i^2 - \alpha_r[n_0 + \delta n(t)][p_0 + \delta p(t)]$$

$$= -\alpha_r[(n_0 + p_0)\delta n(t) + \delta n^2(t)]$$

$$\delta n(\mathrm{t}) = \Delta n \exp(-t\alpha_r(n_0 + p_0) = \Delta n \exp(-\frac{t}{\tau})$$

$$\tau_n = \frac{1}{\alpha_r(n_0 + p_0)}$$

Adding a low, but equal number of carriers to the first equation. We see that these injected carriers eventually recombine, have a concentration in an exponential form

Photoconductivity

$$\delta n = \tau_n g_{\text{op}} \quad \text{and} \quad \delta p = \tau_p g_{\text{op}}$$

$$\Delta\sigma = q g_{\text{op}}(\tau_n \mu_n + \tau_p \mu_p)$$

$$n = n_i e^{(F_n - E_i)/kT}$$

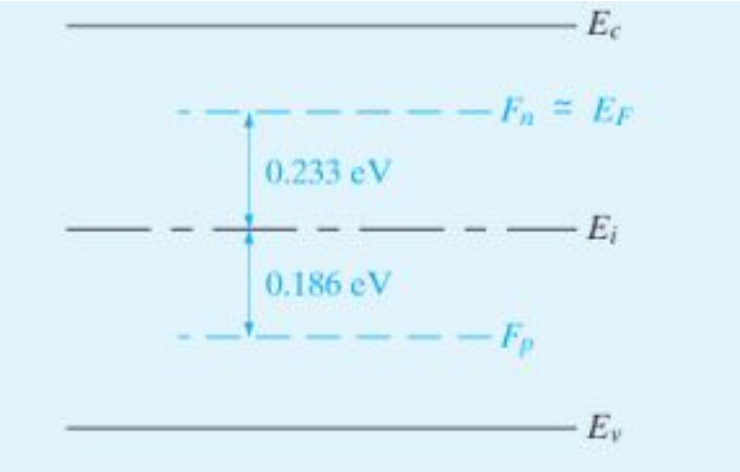
$$p = n_i e^{(E_i - F_p)/kT}$$

Optical Absorption: $I(x) = I_0 \exp(-\alpha x)$

Light transmitted through sample of thickness t: $\mathbf{I}_t = \mathbf{I}_0 e^{-\alpha l}$

Quasi Fermi Levels

- Non equilibrium conditions.
  (Remember!!! Equilibrium: constant T, in the Dark, no applied E-Field)



$$n = n_i e^{(F_n - E_i)/kT}$$

$$p = n_i e^{(E_i - F_p)/kT}$$

Steady State Carrier Injection

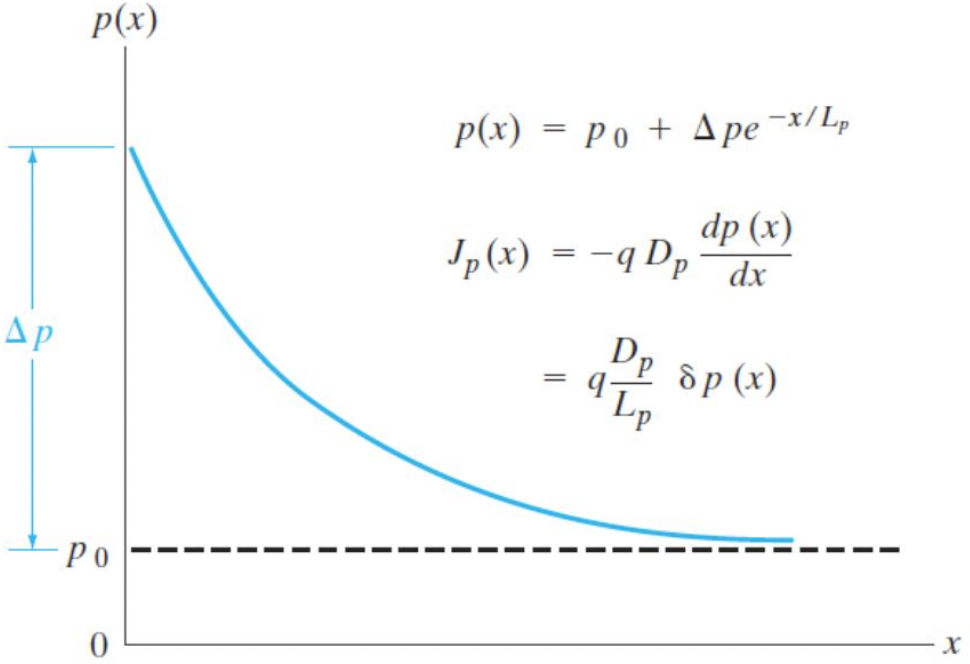
Steady-state ≠ equilibrium

$$\frac{d^2\delta n}{dx^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2}$$

$$\frac{d^2\delta p}{dx^2} = \frac{\delta p}{D_p \tau_p} \equiv \frac{\delta p}{L_p^2}$$

$$L = \sqrt{D\tau}$$

$$\delta p(x) = \Delta p e^{-x/L_p}$$



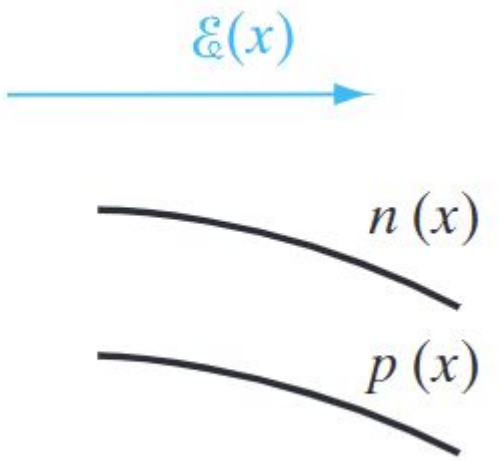
$$p(x) = p_0 + \Delta p e^{-x/L_p}$$

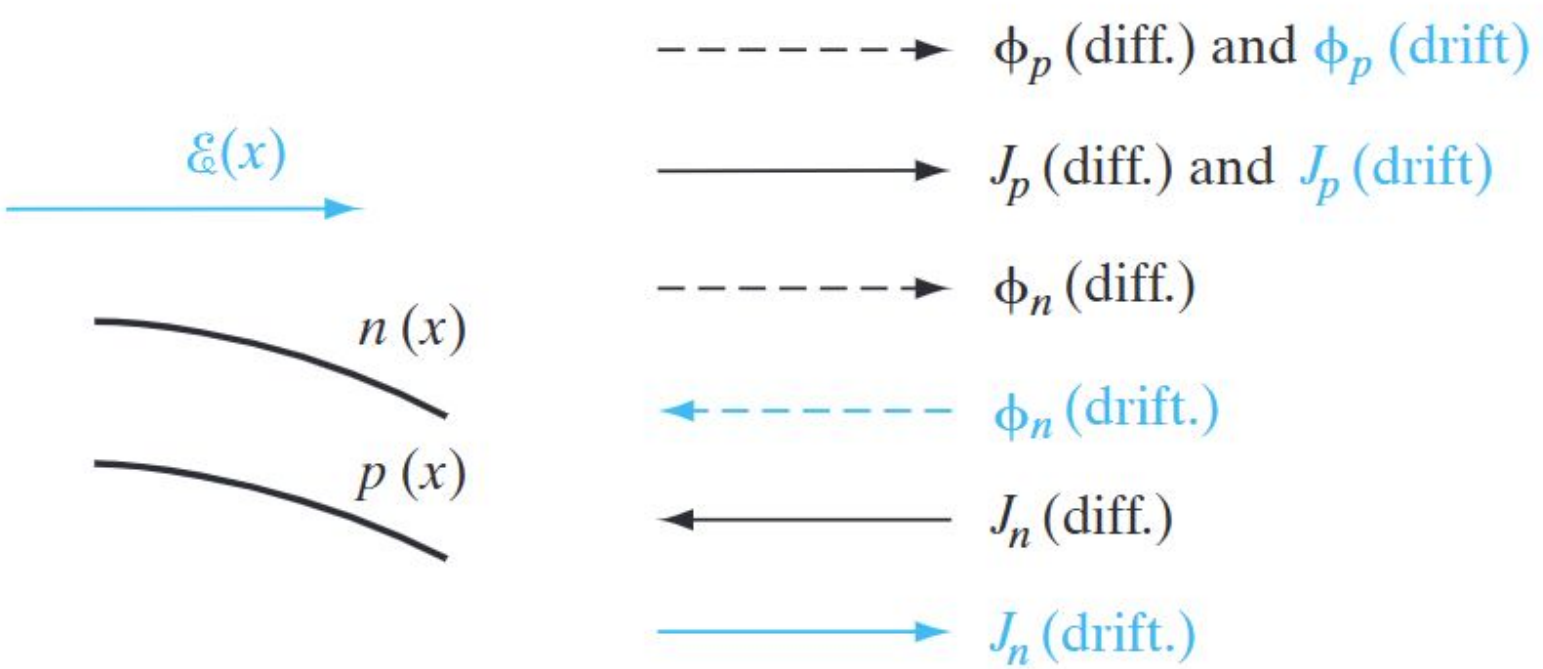
$$J_p(x) = -q D_p \frac{dp(x)}{dx}$$

$$= q\frac{D_p}{L_p} \delta p(x)$$

Practice Problem

What are the directions of the carrier flows from drift? From diffusion?
What are the directions of the currents from drift and diffusion?




$\mathcal{E}(x)$
$n\,(x)$
$p\,(x)$
ϕ_p (diff.) and ϕ_p (drift)
J_p (diff.) and J_p (drift)
ϕ_n (diff.)
ϕ_n (drift.)
J_n (diff.)
J_n (drift.)

Question?

Good luck on the test!

Please fill out the feedback form! →

